



Gadovist® 1.0 mmol/ml

Solution for injection

1. NAME OF THE MEDICINAL PRODUCT

Gadovist 1.0 mmol/ml solution for injection

2. QUALITATIVE AND QUANTITATIVE COMPOSITION

Each ml contains 1.0 mmol gadobutrol (equivalent to 604.72 mg gadobutrol). For full list of excipients, see section 'List of excipients'.

3. PHARMACEUTICAL FORM

Solution for injection.
Clear, colorless to pale yellow solution.

4. CLINICAL PARTICULARS

4.1 Indication(s)

This medicinal product is for diagnostic use only.

Gadovist is indicated in adults, and children of all ages (including term neonates) for

- Contrast enhancement in cranial and spinal magnetic resonance imaging (MRI).

Gadovist is particularly suitable for cases where the exclusion or demonstration of additional pathology may influence the choice of therapy or patient management for detection of very small lesions and for visualization of tumours that do not readily take up contrast media.

Gadovist is also suited for perfusion studies for the diagnosis of stroke, detection of focal cerebral ischaemia and tumour perfusion.

- Contrast enhanced MRI of liver or kidneys in patients with high suspicion or evidence of having focal lesion to classify these lesions as benign or malignant.
- Contrast enhancement in Magnetic Resonance Angiography (CE-MRA).

Gadovist can also be used for MR Imaging of pathologies of the whole body. It facilitates visualisation of abnormal structures or lesions and helps in the differentiation between healthy and pathological tissue.

4.2 Dosage and method of administration

4.2.1 Method of administration

This medicinal product is for intravenous administration only.

For additional instructions see section 'Instructions for use/handling'.

Contrast-enhanced MRI can commence immediately afterwards (shortly after the injection depending on the pulse sequences used and the protocol for the examination). Optimal signal enhancement is observed during arterial first pass for

CE-MRA and within a period of about 15 minutes after injection of Gadovist for other indications (time depending on type of lesion/tissue).

4.2.2 Dosage regimen

Adults:

Dosage depends on indication. A single intravenous injection of 0.1 mmol gadobutrol per kg body weight (equivalent to 0.1 ml Gadovist per kg body weight) is generally sufficient. A total amount of 0.3 mmol gadobutrol per kg body weight (equivalent to 0.3 ml Gadovist per kg body weight) may be administered at maximum.

► Whole Body MRI (except MRA)

In general, the administration of 0.1 ml Gadovist per kg body weight is sufficient to answer the clinical question

► CE-MRA

Imaging of one field of view:

7.5 ml for body weight less than 75 kg

10 ml for body weight of 75 kg or more

(corresponding to 0.1-0.15 mmol per kg body weight)

Imaging of more than one field of view:

15 ml for body weight less than 75 kg

20 ml for body weight of 75 kg or more

(corresponding to 0.2-0.3 mmol per kg body weight)

4.2.3 Special patient populations

► Pediatric population

For children of all ages (including term neonates), the recommended dose is 0.1mmol gadobutrol per kg body weight) for all indications (see section 'Indications').

► Patients with renal impairment

The elimination of gadobutrol is prolonged in patients with renal impairment. However, to ensure diagnostically useful images no dosage adjustment is recommended (see also section 'Special warnings and precautions for use').

4.3 Contraindications

There are no absolute contraindications to the use of Gadovist.

4.4 Special warnings and special precautions for use

Exposure to gadolinium-based contrast agents (GBCAs) increases the risk for Nephrogenic Systemic Fibrosis (NSF) in patients with: acute or chronic severe renal insufficiency (glomerular filtration rate $< 30\text{mL/min/1.73m}^2$), or

► acute renal insufficiency of any severity due to the hepato-renal syndrome or in the perioperative liver transplantation period.

► NSF is a debilitating and sometimes fatal disease affecting the skin, muscle, and internal organs.

Avoid use of GBCAs unless the diagnostic information is essential and not available with non-contrast enhanced magnetic resonance imaging (MRI).

Screen all patients for renal dysfunction by obtaining a history and/or laboratory tests.

When administering a GBCA, do not exceed the dose recommended in product labelling. Allow sufficient time for elimination of the GBCA prior to any readministration.

- Among the factors that may increase the risk for NSF are repeated or higher than recommended doses of a GBCA. - For patients receiving hemodialysis, healthcare professionals may consider prompt hemodialysis following GBCA administration in order to enhance the contrast agent's elimination. However, it is unknown if hemodialysis prevents NSF. - Determine the renal function of patients by obtaining a medical history of conducting laboratory tests that measure renal function prior to using GBCA.
- The risk, if any, for developing NSF among patients with mild to moderate renal insufficiency or normal renal function is unknown.
- Post-marketing reports have identified the development of NSF following single and multiple administrations of GBCAs.

► Risks associated with intrathecal use

Serious, life-threatening and fatal cases, primarily with neurological reactions (e.g. coma, encephalopathy, seizures), have been reported with intrathecal use of gadolinium-based contrast agents (GBCAs). The safety and effectiveness of Gadovist have not been established with intrathecal use. Gadovist is not approved for intrathecal use.

► Hypersensitivity

Particularly careful risk-benefit assessment is required in patients with known hypersensitivity to Gadovist. As with other intravenous contrast agents, Gadovist can be associated with anaphylactoid/hypersensitivity or other idiosyncratic reactions, characterized by cardiovascular, respiratory or cutaneous manifestations, and ranging to severe reactions including shock.

The risk of hypersensitivity reactions is higher in case of:

- previous reaction to contrast media
- history of bronchial asthma - history of allergic disorders

In patients with an allergic disposition the decision to use Gadovist must be made after particularly careful evaluation of the risk-benefit ratio.

Most of these reactions occur within half an hour of administration.

Therefore, post-procedure observation of the patient is recommended.

Medication for the treatment of hypersensitivity reactions as well as preparedness for institution of emergency measures are necessary.

Delayed reactions (after hours up to several days) have been rarely observed (see section 'Undesirable effects').

► Impaired renal function

Prior to administration of Gadovist all patients should be screened for renal dysfunction by obtaining a history and/or laboratory tests.

In patients with severely impaired renal function the benefits must be weighed carefully against the risks, since contrast medium elimination is delayed in such cases.

Because Gadobutrol is renally excreted sufficient period of time for elimination of the contrast agent from the body prior to any re-administration in patients with renal impairment should be ensured. Usually, complete recovery in the urine was seen in patients with mild or moderate renal impairment within 72 hours. In patients with severely impaired renal function at least 80 % of the administered dose was recovered in the urine within 5 days.

Gadovist can be removed from the body by hemodialysis. After 3 dialysis sessions approx. 98 % of the agent are removed from the body. For patients already receiving hemodialysis at the time of Gadovist administration, prompt initiation of hemodialysis following the administration of Gadovist should be considered, in order to enhance the contrast agent's elimination.

There have been reports of nephrogenic systemic fibrosis (NSF) (see section 'Undesirable effects') associated with the use of gadolinium-containing contrast agents including Gadovist in patients with

- acute or chronic severe renal impairment ($\text{GFR} < 30 \text{ ml/min/1.73m}^2$) or
- acute renal insufficiency of any severity due to the hepato-renal syndrome or in the perioperative liver transplantation period.

Therefore, Gadovist should only be used in these patients after careful risk/benefit assessment

► Seizure disorders

As with other gadolinium-chelate-containing contrast media, special precaution is necessary in patients with a low threshold for seizures.

4.5 Interaction with other medicinal products and other forms of interaction

No interactions studies with other medicinal products have been conducted.

4.6 Pregnancy and lactation

Pregnancy

Traces of gadolinium-based contrast agents can cross the placental barrier and lead to fetal exposure.

The potential risks of an abnormal pregnancy outcome are unknown, as adequate and well controlled clinical studies with Gadovist were not conducted in pregnant women.

A retrospective cohort study comparing pregnant women who had a GBCA MRI to pregnant women who did not have an MRI reported a higher occurrence of stillbirths and neonatal deaths in the group receiving GBCA MRI. However, no increased risk of congenital anomalies was observed. Limitations of this study include a lack of comparison with noncontrast MRI and lack of information about the maternal indication for MRI.

These limitations were further addressed in another retrospective cohort study that found no increased risk for fetal or neonatal death or Neonatal Intensive Care Unit (NICU) admission when comparing pregnancies exposed to GBCA MRI and non-contrast MRI.

Cohort studies and case reports on exposure to GBCAs during pregnancy have not reported a clear association between GBCAs and congenital anomalies, fetal and neonatal death or the need for neonatal intensive care unit admission.

Animal studies at clinically relevant doses have not shown reproductive toxicity after repeated administration.

Gadovist should only be used during pregnancy after a careful benefit-risk analysis.

Lactation

It is unknown whether gadobutrol is excreted in human milk.

There is evidence from non-clinical data that gadobutrol is excreted into breast milk in very small amounts (less than 0.1% of the dose intravenously administered) and the absorption via the gastrointestinal tract is poor (about 5% of the dose orally administered were excreted in the urine).

At clinical doses, no effects on the infant are anticipated and Gadovist can be used during breastfeeding.

4.7 Effects on ability to drive or use machines

Not known.

4.8 Undesirable effects

The overall safety profile of Gadovist is based on data from more than 6,300 patients in clinical trials, and from postmarketing surveillance.

The most frequently observed adverse drug reactions ($\geq 0.5\%$) in patients receiving Gadovist are headache, nausea, and dizziness.

The most serious adverse drug reactions in patients receiving Gadovist are cardiac arrest, acute respiratory distress syndrome / pulmonary edema and severe anaphylactoid reactions.

Delayed allergoid or other idiosyncratic reactions (hours later up to several days) have been rarely observed. Most of the undesirable effects were of mild to moderate intensity.

The adverse drug reactions observed with Gadovist are represented in the table below. They are classified according to System Organ Class. The most appropriate MedDRA term is used to describe a certain reaction and its synonyms and related conditions.

Adverse drug reactions from clinical trials are classified according to their frequencies. Frequency groupings are defined according to the following convention: common: $\geq 1/100$ to $< 1/10$; uncommon: $\geq 1/1,000$ to $< 1/100$; rare: $\geq 1/10,000$ to $< 1/1,000$. The adverse drug reactions identified only during post-marketing surveillance, and for which a frequency could not be estimated, are listed under 'not known'.

Within each frequency grouping, undesirable effects are presented in order of decreasing seriousness.

Table 1: Adverse drug reactions reported in clinical trials or during post-marketing surveillance in patients treated with Gadovist

System Organ Class	Common	Uncommon	Rare	Not known
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Immune system disorders		Hypersensitivity / anaphylactoid reaction* [#] (e.g. anaphylactoid shock [§] , circulatory collapse [§] , respiratory arrest [§] , bronchospasm [§] , cyanosis [§] , oropharyngeal swelling [§] , laryngeal edema [§] , hypotension, blood pressure increased [§] , chest pain [§] , urticaria, face edema, angioedema [§] , conjunctivitis [§] , eyelid edema, flushing, hyperhidrosis [§] , cough [§] , sneezing [§] , burning sensation [§] , pallor [§])		
Nervous system disorders	Headache	Dizziness Dysgeusia Paresthesia	Loss of consciousness* Convulsion Parosmia	
Cardiac disorders			Tachycardia Palpitations	Cardiac arrest*
Respiratory, thoracic and mediastinal disorders		Dyspnea*		Acute Respiratory Distress Syndrome (ARDS)* Pulmonary edema*
Gastrointestinal disorders	Nausea	Vomiting	Dry mouth	
Skin and subcutaneous tissue disorders		Erythema Pruritus (including generalized pruritus)		Nephrogenic Systemic Fibrosis (NSF)
System Organ Class	Common	Uncommon	Rare	Not known
		Rash (including generalized, macular, papular, pruritic rash)		
General disorders and administration site conditions		Injection site reaction ⁰ Feeling hot	Malaise Feeling cold	

* There have been reports of life-threatening and/or fatal outcomes from this ADR

[#] None of the individual symptoms ADRs listed under hypersensitivity/anaphylactoid reaction identified in clinical trials reached a frequency greater than rare (except for urticaria)

[§] Hypersensitivity / anaphylactoid reactions identified only during post-marketing surveillance (frequency not known)

⁰ Injection site reactions (various kinds) comprise the following terms: Injection site extravasation, injection site burning, injection site coldness, injection site warmth, injection site erythema or rash, injection site pain, injection site hematoma

4.9 Overdose

Single doses of gadobutrol as high as 1.5 mmol gadobutrol/kg body weight were tolerated well. In case of inadvertent overdosage, cardiovascular monitoring (including ECG) and control of renal function are recommended as a measure of precaution.

5. PHARMACOLOGICAL PROPERTIES

5.1 Pharmacodynamic properties

Pharmacotherapeutic group: Paramagnetic contrast media
ATC-Code: V08C A09

Mechanism of action

Gadovist is a paramagnetic contrast agent for magnetic resonance imaging.

The contrast-enhancing effect is mediated by gadobutrol, a neutral (non-ionic) complex consisting of gadolinium (III) and the macrocyclic ligand dihydroxy-hydroxymethylpropyl-tetraazacyclododecane-triacetic acid (butrol).

When T₁-weighted scanning sequences are used in proton magnetic resonance imaging, the gadolinium ion induced shortening of the spin-lattice relaxation time of excited atomic nuclei leads to an increase of the signal intensity and, hence, to an increase of the image contrast of certain tissues. In T₂*-weighted sequences, however, the induction of local magnetic field inhomogeneities by the large magnetic moment of gadolinium and at high concentrations (during bolus injection) leads to a signal decrease.

Pharmacodynamic effects

Gadobutrol leads to distinct shortening of the relaxation times even in low concentrations due to the pronounced relaxivity. The relaxivity, studied *in vitro* at physiological conditions and clinical relevant field strengths (1.5 and 3.0 T), of gadobutrol is in the range of 4.4 – 5.2 L/mmol/sec (see Table 2).

Table 2: Range of T₁ relaxivities [L/mmol/sec] of GBCAs studied *in vitro* at physiological conditions at 1.5 & 3T

<u>Field Strength (T)</u>	<u>Macrocyclic GBCAs</u>				<u>Linear GBCAs</u>			
	<u>Gado-butrol</u>	<u>Gado-teric acid</u>	<u>Gado-teridol</u>		<u>Gado-pentetate</u>	<u>Gadodiamide</u>	<u>Gadobenate</u>	<u>Gado-xetate</u>
1.5	4.6 – 5.2	3.6 – 3.9	4.1 – 4.3		4.1 – 4.2	4.3 – 4.5	6.2 – 6.3	6.9 – 7.3
3.0	4.4 – 5.0	3.3 – 3.5	3.4 – 3.7		3.5 – 3.7	3.5 – 4.0	5.0 – 5.5	5.4 – 6.2

The macrocyclic ligand forms a stable complex with the paramagnetic gadolinium ion with extremely high *in vivo* and *in vitro* stability (thermodynamic stability constant: log K = 21-22). Gadobutrol is a highly water-soluble, extremely hydrophilic compound with a partition coefficient between n-butanol and buffer at pH 7.6 of about 0.006. The substance does not display any inhibitory interaction with enzymes.

The complex stability of various GBCAs has been studied *in vitro* at physiological conditions. The amounts of released gadolinium ions for gadobutrol as well as for the other macrocyclic agents were below the detection threshold demonstrating the high complex stability of these agents at physiological conditions (see Table 3).

No release of gadolinium ions was observed from the macrocyclic contrast agents gadobutrol, gadoteridol, and gadoterate meglumine during the incubation period of 15 days at 37°C.

Table 3: Gadolinium (Gd) release after 15 days in native human serum (at pH 7.4 and 37 °C) and the initial rate of Gd release determined by HPLC-ICP-MS analysis (95% confidence interval in brackets)

<u>Structural class of GBCA</u>	<u>INN</u>	<u>Gd³⁺ release after 15 days (%)</u>	<u>Initial rate (%/day)</u>
Non-ionic linear	Gadoversetamide	21 (19-22) %	0.44 (0.40-0.51) %/d

	Gadodiamide	20 (17-20) %	0.16 (0.15-0.17) %/d
Ionic linear	Gadopentetate dimeglumine	1.9 (1.2-2.0) %	0.16 (0.12-0.36) %/d
	Gadobenate dimeglumine	1.9 (1.3-2.1) %	0.18 (0.13-0.38) %/d
	Gadofosveset trisodium	1.8 (1.4-1.9) %	0.12 (0.11-0.18) %/d
	Gadoxetate disodium	1.1 (0.8-1.2) %	0.07 (0.05-0.08) %/d
Macrocylic	Gadobutrol	All measurements were below limit of quantification (i.e. < 0.1% after 15 d)	
	Gadoteridol		
	Gadoterate meglumine		

5.2 Pharmacokinetic properties

General introduction

Gadobutrol behaves in the organism like other highly hydrophilic biologically inert, renally excreted compounds (e.g. mannitol or inulin).

Absorption and Distribution

Gadobutrol is rapidly distributed in the extracellular space. Protein binding is negligible. After a dose of 0.1 mmol gadobutrol/kg body weight, an average of 0.59 mmol gadobutrol/l plasma was measured 2 minutes after the injection and 0.3 mmol gadobutrol/l plasma 60 minutes p.i.

Investigations in animals:

In rats, all GBCAs enter the brain via the blood-CSF barrier to a similar and very low extent. In rats it has been demonstrated that GBCAs including gadobutrol do not penetrate the intact blood-brain barrier.

In rabbits the placental transfer was insignificant, 0.01 % of the administered dose being detected in the fetuses.

In lactating rats, less than 0.1% of the total administered dose was excreted into the breast milk. In rats, absorption after oral administration was found to be very small and amounted to about 5 % based on the fraction of the dose excreted in urine.

Enterohepatic circulation has not been observed.

Presence of gadolinium in the brain and body:

After administration of all GBCAs, traces of gadolinium may be detected in the brain, bones, skin, liver, urine and other organs and tissues for an extended period of time. Lower concentrations may be detected with macrocyclic GBCAs such as gadobutrol than with linear GBCAs. Increased signal intensity on non-contrast T1-weighted images within the brain, mainly the globus pallidus and the dentate nucleus, has been observed after multiple IV administrations of primarily linear GBCAs. The clinical relevance of these findings is unknown.

Metabolism

Gadobutrol is not metabolized.

Elimination

Gadobutrol is eliminated from plasma with a mean terminal half-life of 1.81 hours (range 1.33 - 2.13 hours).

Gadobutrol is excreted in an unchanged form via the kidneys. The extrarenal elimination is negligible. Renal clearance of gadobutrol is 1.1 to 1.7 ml/min/kg in healthy subjects and, thus, comparable to the renal clearance of inulin, pointing to the fact that gadobutrol is eliminated by glomerular filtration. More than 50 % of the given dose were excreted within two hours after intravenous administration via the urine. Gadobutrol was completely excreted within 24 hours. Less than 0.1 % was eliminated via the feces.

Linearity/non-linearity

The pharmacokinetics of gadobutrol in humans were dose proportional (e.g. C_{max} , AUC) and dose independent (e.g. V_{ss} , $t_{1/2}$), respectively.

Additional information on special populations

► Pediatric population

Pharmacokinetics of gadobutrol in the pediatric population aged <18 years and in adults are similar (see section 'Dosage and method of administration').

Two single dose phase I/III studies in pediatric patients < 18 years have been performed. The pharmacokinetics were evaluated in 130 pediatric patients aged 2 to < 18 years and in 43 pediatric patients <2 years of age (including full-term newborns).

It was shown that the pharmacokinetic profile of gadobutrol in children of all ages is similar to that in adults, resulting in similar values for AUC, body weight normalized plasma clearance and V_{ss} , as well as elimination half-life and excretion rate.

► Geriatric patients

Due to physiological changes in renal function with age, in elderly healthy volunteers (aged 65 years and above) systemic exposure was increased by approximately 33% (men) and 54% (women) and terminal half-life by approximately 33% (men) and 58% (women). The plasma clearance is reduced by approximately 25% (men) and 35% (women), respectively. The recovery of the administered dose in urine was complete after 24 h in all volunteers and there was no difference between elderly and non-elderly healthy volunteers.

► Patients with renal impairment

In patients with impaired renal function, the serum half-life of gadobutrol is prolonged due to the reduced glomerular filtration.

The mean terminal half-life was prolonged to 5.8 hours in mildly to moderately impaired patients ($80 > CL_{CR} > 30$ ml/min) and further prolonged to 17.6 hours in severely impaired patients not on dialysis ($CL_{CR} < 30$ ml/min).

The mean serum clearance was reduced to 0.49 ml/min/kg in mildly to moderately impaired patients ($80 > CL_{CR} > 30$ ml/min) and to 0.16 ml/min/kg in severely impaired patients not on dialysis ($CL_{CR} < 30$ ml/min).

Complete recovery in the urine was seen in patients with mild or moderate renal impairment within 72 hours. In patients with severely impaired renal function about 80 % of the administered dose was recovered in the urine within 5 days (see also sections 'Dosage and method of administration' and 'Special warnings and precautions for use').

Hemodialysis treatment might be considered necessary if renal function is severely restricted. In patients requiring dialysis, gadobutrol was almost completely removed from serum after the third dialysis.

5.3 Preclinical safety data

Non-clinical data reveal no special hazard for humans based on conventional studies of systemic toxicity, genotoxicity, and contact-sensitizing potential.

► Reproduction toxicity

Repeated intravenous dosing in reproductive toxicology studies caused a retardation of embryonal development in rats and rabbits and an increase in embryoletality in rats, rabbits and monkeys at dose levels being 8 to 16 times (based on body surface area) or 25 to 50 times (based on body weight) above the diagnostic dose in humans. It is not known whether these effects can also be induced by a single administration.

► Local tolerance

Experimental local tolerance studies following a single paravenous, subcutaneous as well as intramuscular application indicated that slight local intolerance reactions could occur at the administration site after inadvertent paravenous administration.

► Safety pharmacology

In preclinical cardiovascular safety pharmacology studies, depending on the dose administered, transient increases in blood pressure and myocardial contractility were observed.

These effects have not been observed in humans.

6. PHARMACEUTICAL PARTICULARS

List of excipients

1 N hydrochloric acid
Calcobutrol sodium
Trometamol
Water for injections

Incompatibilities

In the absence of compatibility studies, this medicinal product must not be mixed with other medicinal products.

Shelf life

Store unopened products below 30 °C

After the vial/bottle has been opened or the prefilled syringe has been prepared for use, Gadovist remains stable for 24 hours at 20 to 25 °C and must be discarded thereafter.

Instructions for use / handling Visual inspection

This medicinal product should be visually inspected before use.

Gadovist should not be used in case of severe discoloration, the occurrence of particulate matter or a defective container.

Vials

Gadovist should only be drawn into the syringe immediately before use.

The rubber stopper should never be pierced more than once.

Any contrast medium solution not used in one examination must be discarded.

Prefilled syringes

Gadovist is ready to use

The prefilled syringe must be taken from the pack and should be prepared for the injection immediately before the examination.

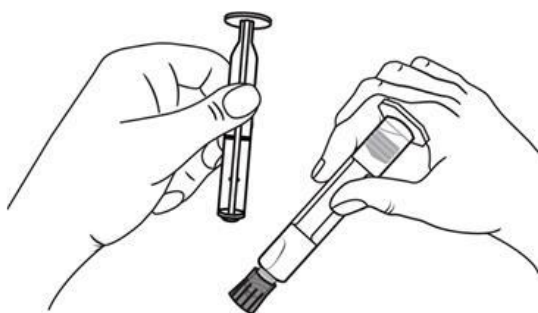
The tip cap should be removed from the prefilled syringe immediately before use. Any contrast medium solution not used in one examination is to be discarded in accordance with local requirements.

The peel-off tracking label on the syringes should be stuck onto the patient record to enable accurate recording of the gadolinium contrast agent used. The dose used should also be recorded.

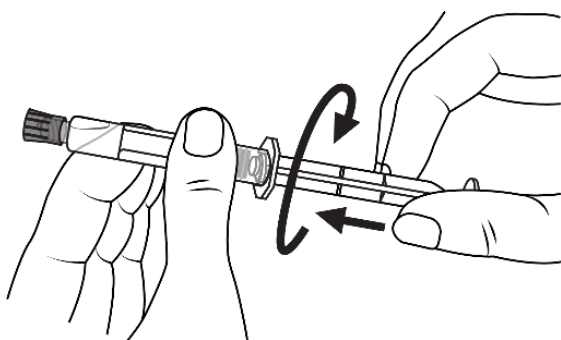
If electronic patient records are used, the name of the product, the batch number and the dose should be entered into the patient record.

Polymer Prefilled Syringes only:

Hand injection



1. Take out the syringe and plunger rod

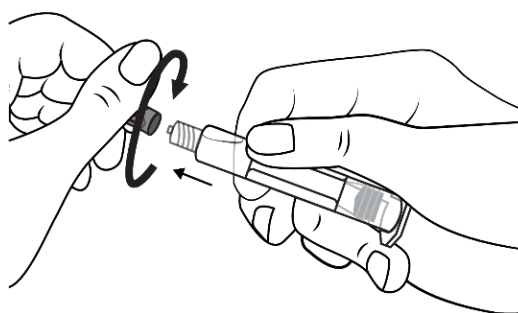


2. Turn the plunger rod clockwise into the syringe

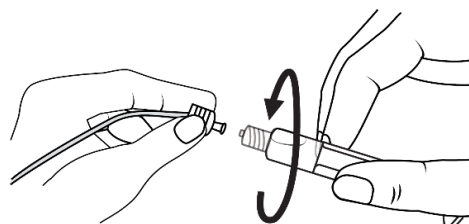
Injection with a power injector

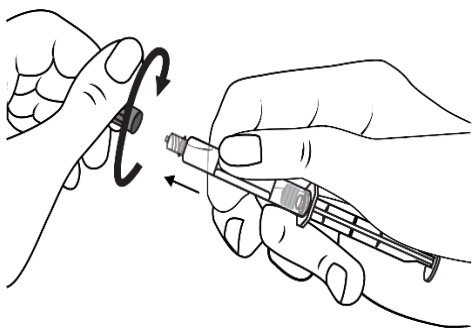


1. Take out the syringe



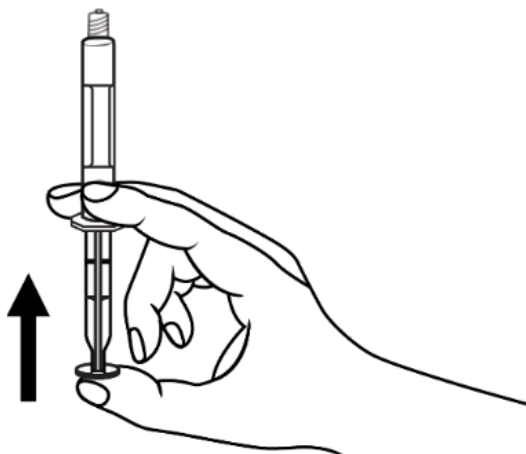
2. Open the cap with a twist





3. Connect the tip of the syringe to the tubing system by turning it clockwise. Proceed according to the instructions for the device

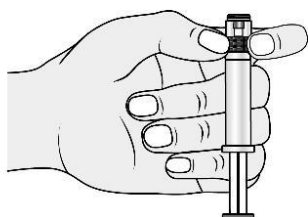
3. Open the cap with a twist



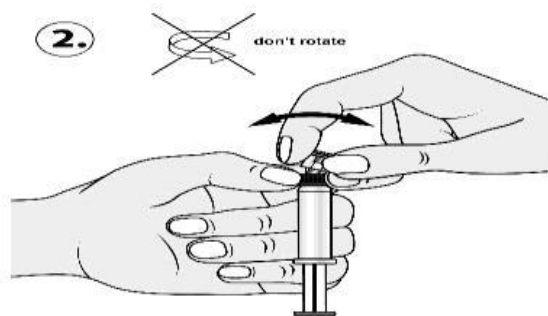
4. Remove the air in the syringe

Glass syringe only:

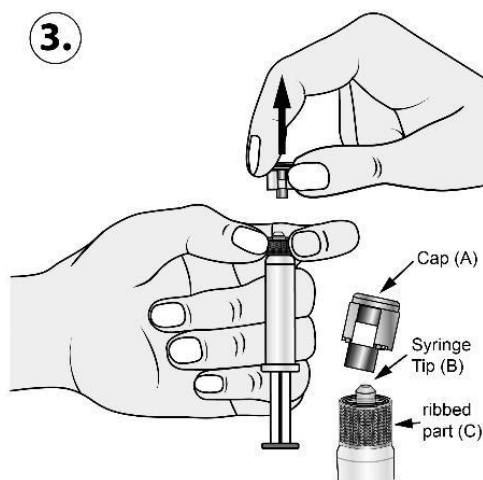
①.



1. Hold the syringe upright on the ribbed part (C) of the closure system



2. With the other hand, take hold of the cap (A) of the closure system and gently tilt back and forth carefully until the cap disconnects and can be pulled off (All seals will be broken).



3. Remove the cap (A) in straight upward direction. Do not touch the syringe tip (B) to keep sterile.

Large volume containers

In addition, the following applies to use of the infusion bottle containing 65 ml:
The contrast medium must only be administered by means of an automatic injector.
Instructions of the device manufacturer must be followed. For further information see also section 'Shelf life'.

Prefilled plastic cartridges

Administration of contrast media should be performed by qualified personnel with the appropriate procedures and equipment.

Sterile technique must be used in all injections involving contrast media.

Instructions of the device manufacturer must be followed.

Any contrast medium solution not used in one examination must be discarded. Presentation

Available as solution for injection (7.5ml, 15ml & 30ml vial) and pre-filled syringe (5ml, 7.5ml, 10ml, 15ml & 20ml) Not all pack sizes and presentations are marketed.

Manufacturer

Bayer AG
Muellerstrasse 170-178
13353 Berlin
Germany

Date of Revision of Text

Apr 2025